Maintenance Platform Verification

AS-P1-RPT-001

Technical Brief

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Student exercise, not for construction.

A 6.0 m × 3.0 m steel maintenance platform was checked using hand calculations, an Excel workbook, and a 3D Robot model. The beam and joist results agreed closely across all three methods. Differences in the post forces and several matters concerning the bracing are documented below for review.

PURPOSE AND SCOPE

I prepared this brief to record the gravity-load checks on the platform framing and compare the results from three separate methods. The work covers bending stress, shear stress, and deflection in the beams and joists, together with axial stress and elastic Euler buckling of the posts.

This is an elastic analysis exercise, not a complete structural design. CSA S16 resistance factors, lateral-torsional buckling, code column resistance, connections, and guard lateral loading have not been checked. The platform is not intended for construction.

DESIGN BASIS

Item	Basis	Item	Basis
Geometry	6.0 m × 3.0 m; top of steel 3,000 mm	Dead load	0.5 kPa grating plus self-weight
B1 main beams	W310×39; simply supported; 6.0 m span	Live load	4.8 kPa (assumed)
J1 joists	W150×22; 1.5 m spacing (5 total); 3.0 m span	Equipment load	10 kN at midspan of each B1 (assumed)
C1 posts	HSS127×127×6.4; 3.0 m; pinned; K = 1.0	Load combinations	ULS: 1.25D + 1.5L; SLS: D + L
BR1 bracing	L64×64×6.4 X-bracing on all four sides	Deflection criterion	L/360 under D + L (assumed)
Steel	CSA G40.21 350W (assumed); E = 200,000 MPa		

Section properties were taken from the CISC section database available through beamdimensions.com. The loads and criteria identified as assumed have not been supported by a cited design source.

VERIFICATION PROCEDURE

I first worked through the gravity load path by hand in Mathcad Prime (AS-P1-CAL-001), starting with the joists, then the main beams, and finally the posts. I repeated the calculations in a parametric Excel workbook (AS-P1-CAL-002). The workbook uses named input cells and includes a CHECK tab comparing the three result sets against a 2% difference criterion.

I then built a 3D frame model in Autodesk Robot Structural Analysis (AS-P1-MOD-001). Member self-weight was included, and the joist end releases were set to release both bending rotations while retaining torsional restraint. I also prepared the AutoCAD drawings (S-101 and S-201) and a Revit model (AS-P1-MOD-002, sheet S-301). The Revit schedules were used to check the modelled sections and prepare a material take-off.

ANALYSIS RESULTS

The moments, shears, and deflections for B1 and J1 were effectively identical in all three methods. The post reaction differed by 1.7%, mainly because the Robot model included brace self-weight. The post member force requires separate consideration, as explained in the findings.

Quantity	Unit	Mathcad	Excel	Robot	Robot vs hand	Limit
J1 factored moment, M_f	kN·m	13.51	13.52	13.51	0.0%	–
J1 factored shear, V_f	kN	18.02	18.02	18.02	0.0%	–
J1 service deflection	mm	3.56	3.56	3.56	0.0%	8.33 mm
B1 factored moment, M_f	kN·m	78.70	78.70	78.69	0.0%	–
B1 factored shear, V_f	kN	35.96	35.96	35.95	0.0%	–
B1 service deflection	mm	10.72	10.72	10.72	0.0%	16.67 mm
C1 factored axial load	kN	46.03	46.03	46.83	+1.7%	–
C1 Euler buckling load, P_{cr}	kN	1,546	1,546	Not run	–	–

The hand-calculated elastic stresses were 143 MPa in bending and 20 MPa in shear for B1; 85 MPa in bending and 20 MPa in shear for J1; and 16 MPa in axial stress for C1. The factored post load is approximately 3% of its Euler buckling load. These elastic stresses are below the assumed yield strength of 350 MPa, but that comparison is not a CSA S16 strength verification. The small difference between the Mathcad and Excel J1 moment values comes from using g = 9.807 m/s^2 and 9.81 m/s^2, respectively.

FINDINGS AND OUTSTANDING ITEMS

The bracing needs the most attention before any further design work. Its slenderness is high, and the current arrangement depends on how the crossing braces are connected and how they are intended to resist load. The findings below are also recorded in the REVIEW_REGISTER tab of AS-P1-MTO-001.

No.	Item	Finding and action	Status
1	C1 reaction	Robot reports 46.83 kN against 46.03 kN by hand. Its model includes approximately 2.6 kN of bracing self-weight, equivalent to roughly 0.8 kN per post at ULS. Difference explained and accepted.	Closed
2	C1 member force	The Robot post member force is 41.9 kN, versus the Robot reaction of 46.83 kN; approximately 4.9 kN reaches the support through the X-bracing in the model. The hand value is retained as the conservative check.	Open
3	BR1 slenderness	Using r_{min} = 12.5 mm calculated from I_{yp} and A, L/r is approximately 537 for the full long-brace length and 268 for the half-length. The full length exceeds both reference limits of 300 in tension and 200 in compression; the half-length satisfies the tension limit but not the compression limit. Proposed action: bolt the braces at the crossing and use a tension-only arrangement; notes added to S-101 and S-201 at R2.	Open
4	BR1 sag	Robot shows approximately 24 mm of self-weight sag in the long braces (about L/280). The proposed connection at the crossing is expected to reduce this, but it has not been verified.	Open
5	Revit model	The volume schedule exposed an incorrect W6X15 brace family, with an area around 2,950 mm^2 rather than the expected 766 mm^2. The type was corrected to the L64$\times$64$\times$6.4 equivalent and the schedule was checked again.	Closed
6	Material take-off	CISC mass-per-metre values give 1,298 kg of members. Revit volumes give 1,327 kg (+2.3%), reflecting imperial-family geometry. Applying an assumed 10% connection allowance to the CISC quantity gives 1,428 kg. The allowance remains to be confirmed.	Closed

QUESTIONS FOR REVIEW

Before revising the analysis, I would appreciate feedback on the following:

1. Is a tension-only X-bracing arrangement with a bolted crossing appropriate for a platform of this size, or would a stiffer section normally be used?
2. Should the gravity load taken by the bracing be included when checking the posts, or is it normally excluded from the design model?
3. Are 4.8 kPa live load and an L/360 service deflection limit reasonable assumptions for an industrial maintenance platform? What source should be used to establish them?
4. Does the guard lateral load need to be carried through the bracing design?
5. Is a 10% allowance for connections reasonable at this stage of a material take-off?
6. What additional checks and detailing would be needed to develop this exercise into a CSA S16-compliant design?

DOCUMENTS ISSUED

Document	Description	Rev.	Document	Description	Rev.
AS-P1-CAL-001	Mathcad hand calculations	R0	S-101, S-201	AutoCAD plan, sections and notes	R2
AS-P1-CAL-002	Excel calculations and three-way comparison	R0	AS-P1-MOD-002, S-301	Revit model, schedules and sheet	R0
AS-P1-MOD-001	Robot 3D model and result plots	R0	AS-P1-MTO-001	Material take-off, slenderness check and review register	R0